

Worksheet — Lines and Angles (Class 6 CBSE)

Total marks: **25** · Questions: **11**

Instructions: Attempt all 10 questions. Use a protractor where needed. Show working for the subjective questions. Total time: 30 minutes.

Q1. An angle is formed by: [1 mark]

- (A) two parallel lines
- (B) two rays sharing a common endpoint
- (C) a circle and a line
- (D) a triangle

Q2. Vertically opposite angles formed by two intersecting lines are always equal. [1 mark]

- (A) True
- (B) False

Q3. An angle of 90° is called a ____ angle. [1 mark]

Q4. Classify an angle of 135° . [1 mark]

- (A) acute
- (B) right
- (C) obtuse
- (D) straight

Q5. Two lines intersect; one angle is 70° . What is the angle directly opposite? [2 marks]

- (A) 70°
- (B) 90°
- (C) 110°
- (D) 180°

Q6. Explain in your own words the difference between a line, a line segment, and a ray. [2 marks]

Q7. An angle has vertex P with arms going through K and L. List all valid names for this angle. [2 marks]

Q8. Classify each angle: (i) 35° , (ii) 90° , (iii) 165° , (iv) 230° . [3 marks]

Q9. Two lines intersect; one angle is 40° . Find the other three angles, naming the rule used. [3 marks]

Q10. Two lines intersect. The four angles around the intersection are labelled in order $\angle 1$, $\angle 2$, $\angle 3$, $\angle 4$. Given $\angle 1 = 65^\circ$, find each of $\angle 2$, $\angle 3$, $\angle 4$ and state the rule used for each. [5 marks]

Q11. Find three angles around your home or classroom. Sketch each (or describe) and classify it as acute, right, obtuse, straight or reflex. Bonus: estimate the measure in degrees. [4 marks]

Answer Key

Worksheet — Lines and Angles (Class 6 CBSE)

Q1. An angle is formed by:

Answer: two rays sharing a common endpoint

Two rays meeting at a common vertex form an angle.

MCQ · EASY · 6-CBSE-MATH-LA-02

Q2. Vertically opposite angles formed by two intersecting lines are always equal.

Answer: True

Standard rule.

TRUE_FALSE · EASY · 6-CBSE-MATH-LA-07

Q3. An angle of 90° is called a ____ angle.

Answer: right

90° = right angle.

FILL_BLANK · EASY · 6-CBSE-MATH-LA-05

Q4. Classify an angle of 135° .

Answer: obtuse

135° is between 90° and 180° .

MCQ · EASY · 6-CBSE-MATH-LA-05

Q5. Two lines intersect; one angle is 70° . What is the angle directly opposite?

Answer: 70°

Vertically opposite angles are equal.

MCQ · MEDIUM · 6-CBSE-MATH-LA-07

Q6. Explain in your own words the difference between a line, a line segment, and a ray.

Answer: A line extends without end in both directions and has no endpoints. A line segment is a piece of a line with two endpoints and a definite length. A ray has one endpoint and extends without end in the other direction.

Number of endpoints distinguishes them.

SHORT_ANSWER · EASY · 6-CBSE-MATH-LA-01

Q7. An angle has vertex P with arms going through K and L. List all valid names for this angle.

Answer: $\angle KPL$, $\angle LPK$, and $\angle P$ (since only one angle has vertex P). The vertex P MUST be the middle letter when using three letters.

Vertex in the middle.

SHORT_ANSWER · MEDIUM · 6-CBSE-MATH-LA-02

Q8. Classify each angle: (i) 35° , (ii) 90° , (iii) 165° , (iv) 230° .

Answer: (i) 35° = acute. (ii) 90° = right. (iii) 165° = obtuse. (iv) 230° = reflex.

Classification by measure.

SHORT_ANSWER · EASY · 6-CBSE-MATH-LA-05

Q9. Two lines intersect; one angle is 40° . Find the other three angles, naming the rule used.

Answer: Vertically opposite angle = 40° (vertically opposite are equal). Linear-pair partners = $180^\circ - 40^\circ = 140^\circ$ each. Other three: 40° , 140° , 140° .

Two key rules: vertically opposite equal; linear pair sums to 180° .

SHORT_ANSWER · MEDIUM · 6-CBSE-MATH-LA-07

Q10. Two lines intersect. The four angles around the intersection are labelled in order $\angle 1$, $\angle 2$, $\angle 3$, $\angle 4$. Given $\angle 1 = 65^\circ$, find each of $\angle 2$, $\angle 3$, $\angle 4$ and state the rule used for each.

Answer: $\angle 2$ (adjacent to $\angle 1$) = $180^\circ - 65^\circ = 115^\circ$ (linear pair). $\angle 3$ (opposite $\angle 1$) = 65° (vertically opposite). $\angle 4$ (adjacent to $\angle 1$) = $180^\circ - 65^\circ = 115^\circ$ (linear pair). All four sum to 360° .

Practises the rule used at each step.

LONG_ANSWER · HARD · 6-CBSE-MATH-LA-07

Q11. Find three angles around your home or classroom. Sketch each (or describe) and classify it as acute, right, obtuse, straight or reflex. Bonus: estimate the measure in degrees.

Answer: Example answers: (1) Corner of a notebook $\rightarrow 90^\circ$ (right). (2) An open door at 45° from its frame $\rightarrow$ acute. (3) Hands of a clock at 4:00 $\rightarrow$ about 120° (obtuse). Many correct answers; rubric checks each angle has a sketch/description, classification, and a reasonable degree estimate.

Encourages real-world observation.

LONG_ANSWER · HARD · 6-CBSE-MATH-LA-08